

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. (ME, CL) Examination
November 2010

CL 202 Strength of Materials

FOR REFERENCE

Date: 27.11.2010, Saturday

Time: 10:30 a.m. To 01:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.



SECTION - I

- Q - 1 (a) Derive the shear stress equation for rectangular cross-section and show that $q_{\max} = \frac{3}{2} q_{\text{avg}}$. 03
- (b) An I-section member has dimensions as shown in Figure 1. The member is used as a simply supported beam of span 4.0 m. Calculate the uniformly distributed load which can be applied over the entire span length such that maximum shearing stress induced in the cross-section is not to exceed 4.0 MN/m^2 . Find out the maximum bending stress induced in the beam? 04
- Q - 2 (a) A triangular section of a beam has base width of 30 cm and altitude of 25 cm. The section is subjected to a shear force of 150 kN. Determine: (i) average shear stress (ii) maximum shear stress and (iii) shear stress at the Neutral axis. Sketch the shear stress distribution diagram along the depth of section. 03
- (b) Derive equation of torsion with usual notations. 04
- (c) A solid shaft has to transmit 90 kW power at 160 rpm. Find suitable diameter for the shaft if maximum torque transmitted in each revolution exceeds the mean by 25%. Take $f_s = 70 \text{ MPa}$. 03
- (d) A closed coiled helical spring has mean diameter of 120 mm. The diameter of wire is 10 mm and having 9 complete turns. It is subjected to an axial twist of 3 N-m. Calculate the increase in number of turns and bending stress induced in the section. Assume $E = 2 \times 10^5 \text{ N/mm}^2$. 04

OR

- Q - 2 (a) The section of a beam shown in Figure 2 is subjected to a shear force of 200 kN. Determine the shear stress at significant points and sketch the shear stress distribution diagram. 03
- (b) Calculate the maximum intensity of shear stress induced and angle of twist in degrees for a solid shaft of length 10 m and 100 mm diameter, transmitting 112.5 kW at 150 rpm. Take $N = 8.2 \times 10^4 \text{ MPa}$. 04
- (c) A solid circular shaft of steel is 50 mm in diameter. Find the power transmitted at 120 rpm, if the permissible shear stress is 62.5 MPa. 03
- (d) A shaft of 100 mm diameter transmits 112.5 kW power at 180 rpm. A flanged coupling is keyed to the shaft by means of a key 100 mm long and 30 mm wide. The coupling has 6 bolts of 20 mm diameter symmetrically arranged along the bolt circle of 300 mm diameter. Calculate shearing stresses in the shaft, the key and the bolts of the coupling. 04
- Q - 3 Attempt any Two 14

- (a) At a point in a strained material, the intensities of Normal stress across two planes at right angles to each other are as shown in Figure 3. Locate the principal planes and evaluate the principal stresses (p_1 & p_2). Also find the planes of maximum shear stress ($p_{r \max}$) & its location. Find Normal (p_n), Tangential (p_t) stresses on a plane inclined at 30° with vertical as

shown in Figure.3. Use both Analytical and graphical method.

- (b) A point in a strained material is subjected to the stresses as shown in **Figure 4**. Find Normal (p_n), Tangential (p_t) and Resultant (p_r) stresses on a plane inclined at 50° with vertical as shown in Figure 4. Also find maximum Tangential stress ($p_{t \max}$). Use both Analytical and graphical method.
- (c) At a point in a strained material, the intensities of Normal stress across two planes right angles to each other as shown in **Figure 5**. Locate the principal planes and evaluate the principal Stresses (p_1 & p_2). Also find the planes of maximum shear stress ($p_{t \max}$) & its location. Find Normal (p_n), Tangential (p_t) stresses on a plane inclined at 60° with vertical as shown in Figure.5. Use both Analytical and graphical method.

SECTION - II

- Q - 4 (a)** State the assumptions made in the theory of simple bending. 03
- (b)** A flitched beam consists of two timber sections of 150 mm * 300 mm and a steel section 10 mm * 250 mm placed symmetrically between wood sections. Find the moment of resistance of beam if the maximum safe stress in timber is limited to 20 N/mm². What will be the maximum stress in steel if the span of the beam is 5 m. Find the intensity of uniformly distributed load that the beam may carry safely? Take $E_{st} = 200 \text{ kN/mm}^2$ and $E_w = 10 \text{ kN/mm}^2$. 04
- Q - 5 (a)** A simply supported beam of span 10 m and of a rectangular cross-section of 250 mm width and 450 mm depth is subjected to a uniformly distributed load. If the bending stress is not to exceed 120 N/mm². Find (i) its moment of resistance, (ii) maximum permissible intensity of uniformly distributed load. 03
- (b)** A rectangular hole is made in a triangular section as shown in **Figure 6**. Determine the moment of inertia of the section about XX-axis passing through its center of gravity and parallel to the BC. Calculate also the moment of inertia with respect to the base BC. 04
- (c)** State and prove the parallel axis theorem. 03
- (d)** Determine the moment of inertia of the shaded area shown by section lines in the **Figure 7** about centroidal axes XX. 04

OR

- Q - 5 (a)** Derive: $\frac{M}{I} = \frac{\sigma}{y} = \frac{E}{R}$, with usual notations. 03
- (b)** Determine the moment of inertia of the cross-section of an iron beam as shown in **Figure 8** with respect to the centroidal axis parallel to the base AB. 04
- (c)** Find the moment of inertia of the section as shown in **Figure 9** about LM. 03
- (d)** Determine the moment of inertia of the shaded area as shown in **Figure 10** about XX axis. 04
- Q - 6 Attempt any Two** 14
- (a)** Draw Shear Force and Bending Moment Diagram for the beam shown in **Figure 11**.
- (b)** Draw Shear Force and Bending Moment Diagram for the beam shown in **Figure 12**. Also locate the point of maximum bending moment.
- (c)** Draw Shear Force and Bending Moment Diagram for the beam shown in **Figure 13**.



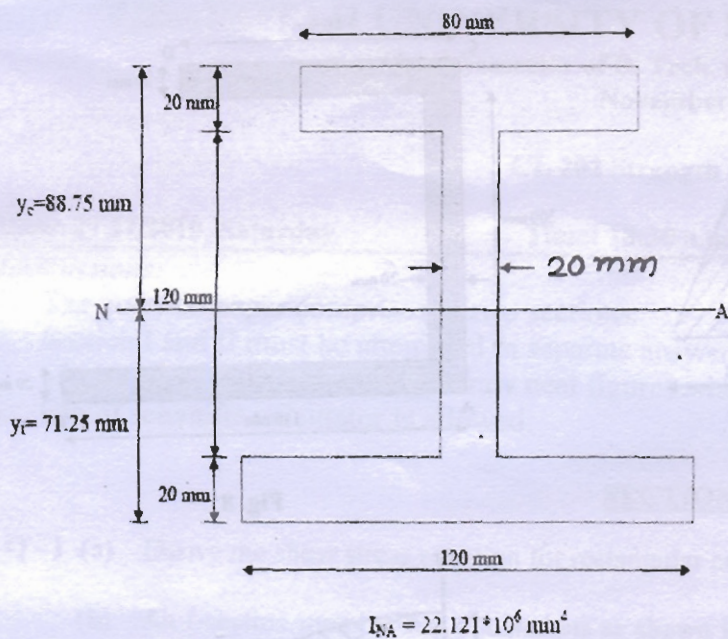


Fig. 1

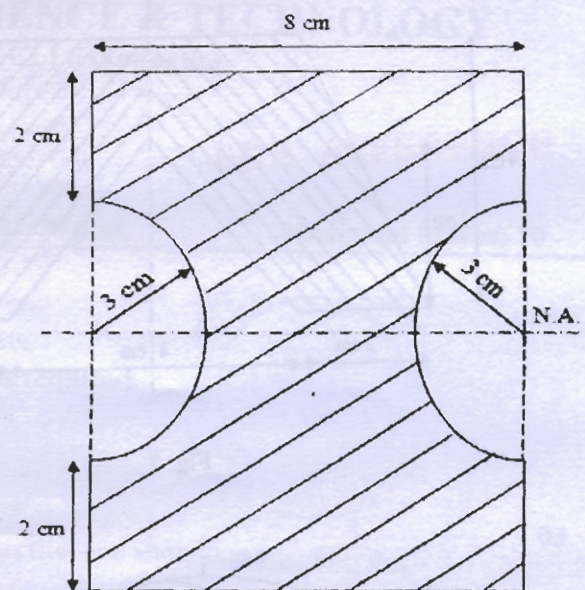


Fig. 2

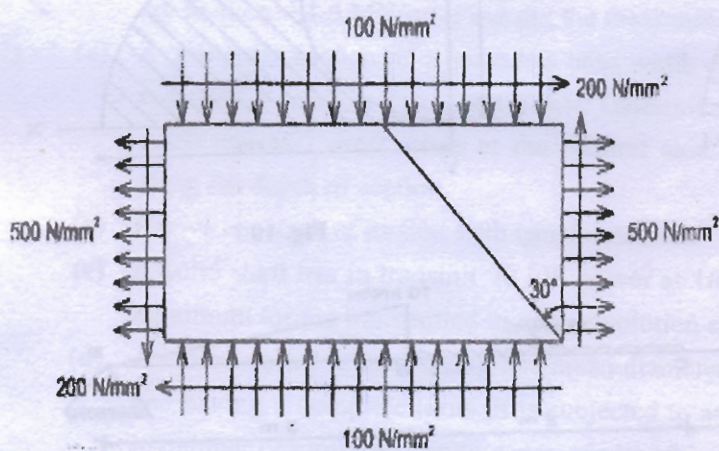


Fig. 3

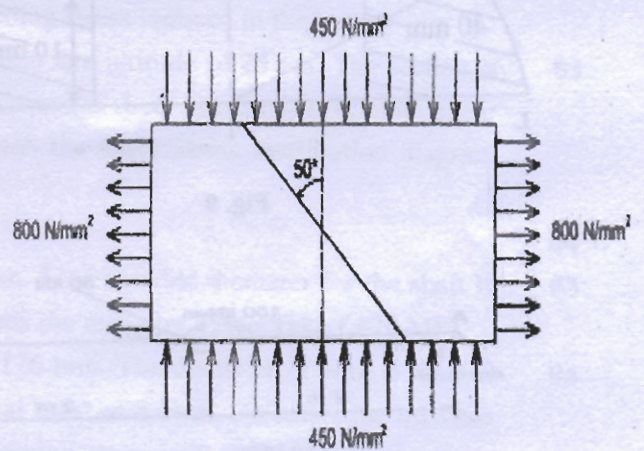


Fig. 4

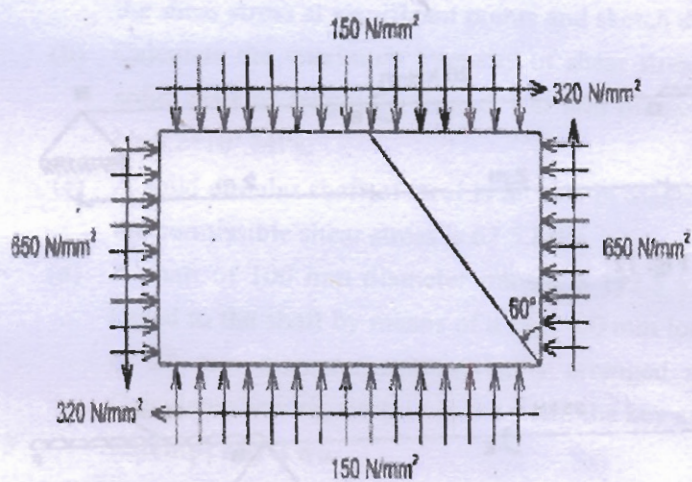


Fig. 5

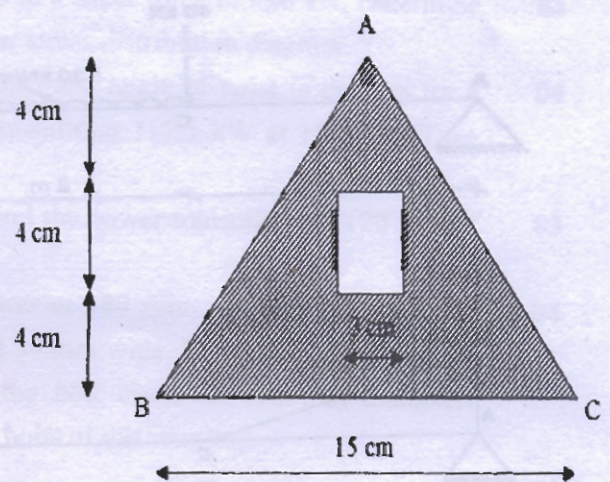


Fig. 6



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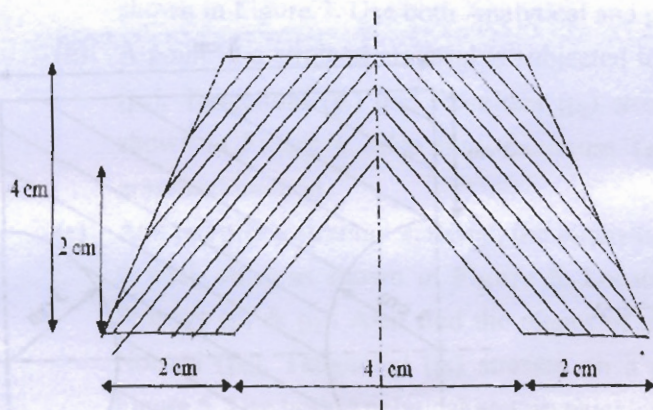


Fig. 7

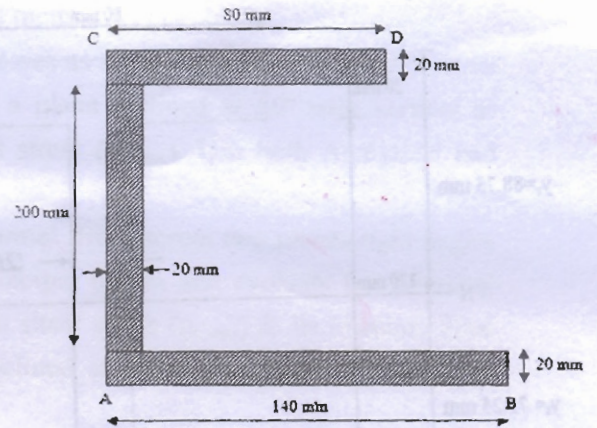


Fig. 8

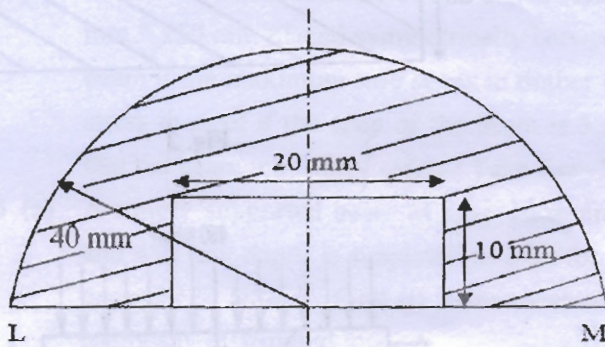


Fig. 9

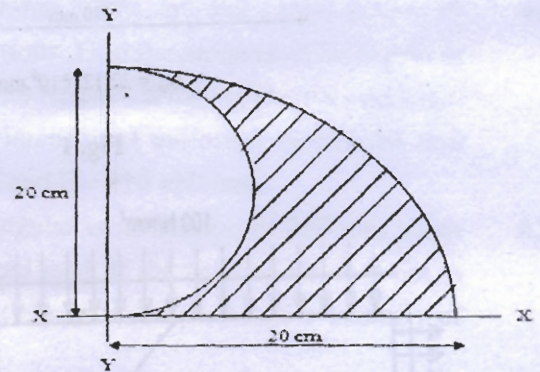


Fig. 10

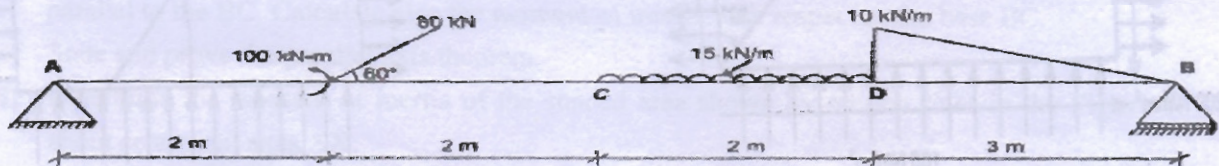


Fig. 11

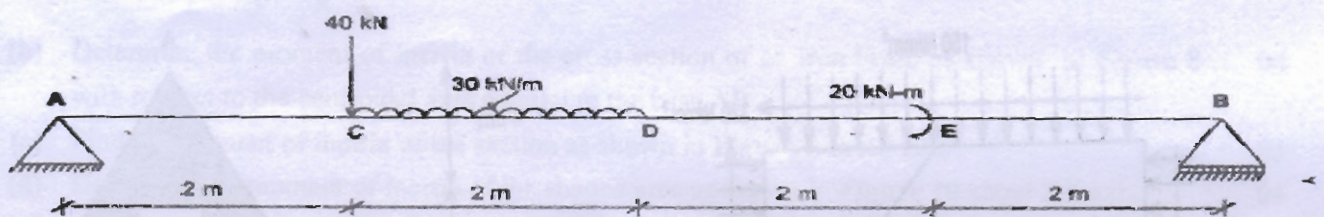


Fig. 12

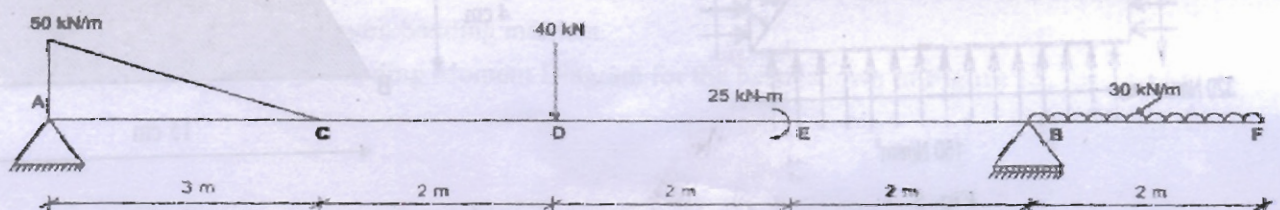


Fig. 13

